4(a)(i)	loss of energy	**B1**
4(a)(ii)	amplitude (of oscillations) decreases (with time)	**B1**
4(b)(i)	$\omega = 2\pi / T$	**C1**
	$T = 0.80$ s, so $\omega = 2\pi / 0.80$ $\omega = 7.9$ rad s^{-1}	**A1**
4(b)(ii)	$\omega^2 = 2k / M$	**C1**
	$7.9^2 = 2k / 1.2$ $k = 37$ N m^{-1}	**A1**
4(c)(i)	(one) smooth curve, not touching the f-axis, with two concave sides meeting at a peak in between them	**B1**
	(one) peak at 1.0ω	**B1**
4(c)(ii)	• lower peak/(whole) line is lower • flatter peak/peak is less sharp • peak at (slightly) lower angular frequency/peak moves to left *any two points, one mark each*	**B2**